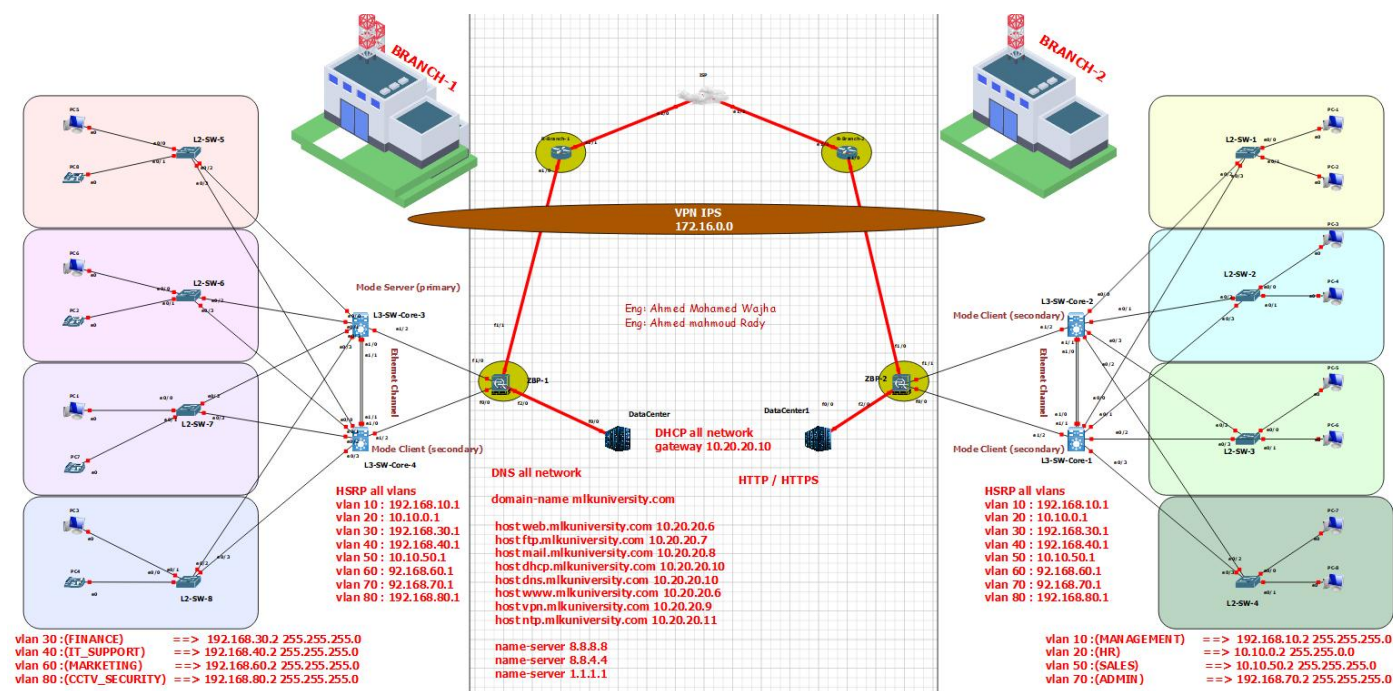




## R-Branch-1 (R-1)

```
hostname R-Branch-1
interface e1/1
description *** Link to ISP ***
ip address 20.20.20.2 255.255.255.252
no shutdown
exit
interface e1/0
description *** Link to ZBP-1 fa1/0 ***
ip address 105.100.50.1 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 1.1.1.1 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.1
network 20.20.20.0 0.0.0.3 area 0
network 105.100.50.0 0.0.0.3 area 0
network 1.1.1.1 0.0.0.0 area 0
default-information originate
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 4
login authentication default
transport input ssh telnet
privilege level 15
exit
```

```
line con 0
login authentication default
password cisco123
exit
interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source 105.100.50.1
tunnel destination 205.200.100.1
ip ospf 1 area 0
no shutdown
exit
ip access-list extended VPN-TRAFFIC
permit ip 10.10.0.0 0.0.255.255 10.11.0.0 0.0.255.255
exit
crypto isakmp policy 10
encr aes 256
hash sha
authentication pre-share
group 14
exit
crypto isakmp key cisco123 address 205.200.100.1
crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac
mode tunnel
exit
crypto map VPN-MAP 10 ipsec-isakmp
set peer 205.200.100.1
set transform-set VPN-SET
match address VPN-TRAFFIC
exit
interface e1/0
crypto map VPN-MAP
exit

end
Wr
```

---

## ISP Router

```
hostname ISP
interface e1/0
description *** Link to R-Branch-1 ***
ip address 20.20.20.1 255.255.255.252
no shutdown
exit
interface e1/1
description *** Link to R-Branch-2 ***
ip address 30.30.30.1 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 2.2.2.2 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.2
```

```
network 20.20.20.0 0.0.0.3 area 0
network 30.30.30.0 0.0.0.3 area 0
network 2.2.2.2 0.0.0.0 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
Exit

end
Wr
```

---

## **R-Branch-2 (R-3)**

```
hostname R-Branch-2
interface e1/2
description *** Link to ISP ***
ip address 30.30.30.2 255.255.255.252
no shutdown
exit
interface e1/0
description *** Link to ZBP-2 fa1/0 ***
ip address 205.200.100.1 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 3.3.3.3 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.3
network 30.30.30.0 0.0.0.3 area 0
network 205.200.100.0 0.0.0.3 area 0
network 3.3.3.3 0.0.0.0 area 0
default-information originate
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
```

```
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
tunnel source 205.200.100.1
tunnel destination 105.100.50.1
ip ospf 1 area 0
no shutdown
exit
ip access-list extended VPN-TRAFFIC
permit ip 10.11.0.0 0.0.255.255 10.10.0.0 0.0.255.255
exit
crypto isakmp policy 10
encr aes 256
hash sha
authentication pre-share
group 14
exit
crypto isakmp key cisco123 address 105.100.50.1
crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac
mode tunnel
exit
crypto map VPN-MAP 10 ipsec-isakmp
set peer 105.100.50.1
set transform-set VPN-SET
match address VPN-TRAFFIC
exit
interface e1/0
crypto map VPN-MAP
exit

end
wr
```

---

## **ZBP-1 (Zone Border Router - Branch-1)**

```
hostname ZBP-1
interface fa1/1
description *** Link to R-Branch-1 e1/0 ***
ip address 105.100.50.2 255.255.255.252
no shutdown
exit
interface fa2/0
description *** Link to DHCP-1 ***
ip address 10.20.20.1 255.255.255.252
no shutdown
```

```
exit
interface fa1/0
description *** Link to L3-SW-Core-3 e1/2 ***
ip address 10.20.20.33 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 4.4.4.4 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.4
network 105.100.50.0 0.0.0.3 area 0
network 10.20.20.0 0.0.0.3 area 0
network 10.20.20.32 0.0.0.3 area 0
network 4.4.4.4 0.0.0.0 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit

end
wr
```

---

## **ZBP-2 (Zone Border Router - Branch-2)**

```
hostname ZBP-2
interface fa1/0
description *** Link to R-Branch-2 e1/0 ***
ip address 205.200.100.2 255.255.255.252
no shutdown
exit
interface fa2/0
description *** Link to DHCP-2 ***
ip address 10.20.20.5 255.255.255.252
no shutdown
exit
interface fa1/1
description *** Link to L3-SW-Core-2 e1/2 ***
ip address 10.20.20.41 255.255.255.252
no shutdown
```

```
exit
interface loopback 0
ip address 5.5.5.5 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.5
network 205.200.100.0 0.0.0.3 area 0
network 10.20.20.4 0.0.0.3 area 0
network 10.20.20.40 0.0.0.3 area 0
network 5.5.5.5 0.0.0.0 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password cisco123
exit

end
wr
```

---

## Data Center (Branch-1)

```
hostname DC-1
interface f0/0
description *** Link to ZBP-1 fa0/0 ***
ip address 10.20.20.2 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 10.20.20.10 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.6
network 10.20.20.0 0.0.0.3 area 0
network 10.20.20.10 0.0.0.0 area 0
exit
ip dhcp excluded-address 10.10.0.1 10.10.0.10
ip dhcp excluded-address 10.10.50.1 10.10.50.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
```

```
ip dhcp excluded-address 192.168.70.1 192.168.70.10
ip dhcp excluded-address 192.168.80.1 192.168.80.10
ip dhcp pool HQ-LAN
network 10.10.0.0 255.255.0.0
default-router 10.10.0.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool HQ-VLAN30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool HQ-VLAN40
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool HQ-VLAN60
network 192.168.60.0 255.255.255.0
default-router 192.168.60.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool HQ-VLAN80
network 192.168.80.0 255.255.255.0
default-router 192.168.80.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool HQ-WLAN
network 10.10.50.0 255.255.255.0
default-router 10.10.50.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 1
exit
ip dns server
ip domain-lookup
ip domain-name mlkuniversity.com
ip host web.mlkuniversity.com 10.20.20.6
ip host ftp.mlkuniversity.com 10.20.20.7
ip host mail.mlkuniversity.com 10.20.20.8
ip host dhcp.mlkuniversity.com 10.20.20.10
ip host dns.mlkuniversity.com 10.20.20.10
ip host www.mlkuniversity.com 10.20.20.6
ip host vpn.mlkuniversity.com 10.20.20.9
ip host ntp.mlkuniversity.com 10.20.20.11
```

```
ip name-server 8.8.8.8
ip name-server 8.8.4.4
ip name-server 1.1.1.1
ip http server
ip http secure-server
ip http authentication local
ip http access-class 10
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 10.10.0.0 0.0.255.255
access-list 10 permit 10.11.0.0 0.0.255.255
access-list 10 deny any log
ip http secure-port 443
ip http secure-ciphersuite high
ip http timeout-policy idle 600 life 86400 requests 10000
username admin privilege 15 secret 123

end
wr
```

---

## **Data Center (Branch-2)**

```
hostname DC-2
interface f0/0
description *** Link to ZBP-2 fa0/0 ***
ip address 10.20.20.6 255.255.255.252
no shutdown
exit
interface loopback 0
ip address 10.20.20.10 255.255.255.255
no shutdown
exit
router ospf 1
router-id 10.10.10.7
network 10.20.20.4 0.0.0.3 area 0
network 10.20.20.10 0.0.0.0 area 0
exit
ip dhcp excluded-address 10.11.0.1 10.11.0.10
ip dhcp excluded-address 10.11.50.1 10.11.50.10
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
ip dhcp excluded-address 192.168.70.1 192.168.70.10
ip dhcp excluded-address 192.168.80.1 192.168.80.10
ip dhcp pool BRANCH-LAN
network 10.11.0.0 255.255.0.0
default-router 10.11.0.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool BRANCH-VLAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 10.20.20.10
```



```
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool BRANCH-VLAN50
network 10.11.50.0 255.255.255.0
default-router 10.11.50.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dhcp pool BRANCH-VLAN70
network 192.168.70.0 255.255.255.0
default-router 192.168.70.1
dns-server 10.20.20.10
domain-name mlkuniversity.com
lease 7
exit
ip dns server
ip domain-lookup
ip domain-name mlkuniversity.com
ip host web.mlkuniversity.com 10.20.20.6
ip host ftp.mlkuniversity.com 10.20.20.7
ip host mail.mlkuniversity.com 10.20.20.8
ip host dhcp.mlkuniversity.com 10.20.20.10
ip host dns.mlkuniversity.com 10.20.20.10
ip host www.mlkuniversity.com 10.20.20.6
ip host vpn.mlkuniversity.com 10.20.20.9
ip host ntp.mlkuniversity.com 10.20.20.11
ip name-server 8.8.8.8
ip name-server 8.8.4.4
ip name-server 1.1.1.1
ip http server
ip http secure-server
ip http authentication local
ip http access-class 10
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 10.10.0.0 0.0.255.255
access-list 10 permit 10.11.0.0 0.0.255.255
access-list 10 deny any log
ip http secure-port 443
ip http secure-ciphersuite high
ip http timeout-policy idle 600 life 86400 requests 10000
username admin privilege 15 secret 123

end
wr
```

---

### **L3-SW-Core-3**

```
hostname L3-SW-Core-3
vtp domain MLK-University
vtp mode server
vtp version 2
vtp password cisco123
```

```
vlan 10
name MANAGEMENT
vlan 20
name HR
vlan 30
name FINANCE
vlan 40
name IT_SUPPORT
vlan 50
name SALES
vlan 60
name MARKETING
vlan 70
name ADMIN
vlan 80
name CCTV_SECURITY
vlan 199
name BLACKHOLE
exit
ip routing
interface range e1/0 , e1/1
channel-group 1 mode active
no shutdown
exit
interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface e1/2
no switchport
ip address 10.20.20.34 255.255.255.252
no shutdown
exit
interface range e0/0 , e0/1 , e0/2 , e0/3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface vlan 10
ip address 192.168.10.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 20
ip address 10.10.0.2 255.255.0.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 30
ip address 192.168.30.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
```

```
exit
interface vlan 40
ip address 192.168.40.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 50
ip address 10.10.50.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 60
ip address 192.168.60.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 70
ip address 192.168.70.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 80
ip address 192.168.80.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 110
standby 1 preempt
exit
```

```
interface vlan 20
standby 1 ip 10.10.0.1
standby 1 priority 110
standby 1 preempt
exit
```

```
interface vlan 30
standby 1 ip 192.168.30.1
standby 1 priority 110
standby 1 preempt
exit
```

```
interface vlan 40
standby 1 ip 192.168.40.1
standby 1 priority 110
standby 1 preempt
exit
```

```
interface vlan 50
standby 1 ip 10.10.50.1
standby 1 priority 110
standby 1 preempt
exit
```

```
interface vlan 60
standby 1 ip 192.168.60.1
standby 1 priority 110
```

```
standby 1 preempt
exit
interface vlan 70
standby 1 ip 192.168.70.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 80
standby 1 ip 192.168.80.1
standby 1 priority 110
standby 1 preempt
exit
router ospf 1
router-id 1.1.1.1
network 10.20.20.32 0.0.0.3 area 0
network 10.10.0.0 0.0.255.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.10.50.0 0.0.0.255 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
aaa accounting commands 15 default start-stop group tacacs+
username admin privilege 15 secret 123
username manager privilege 10 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
line vty 0 2
login authentication default
transport input ssh telnet
privilege level 15
exit
line vty 3 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit
ip access-list standard MGMT-ACL
permit 192.168.10.0 0.0.0.255
deny any log
exit
line vty 0 4
```

```
access-class MGMT-ACL in
exit
snmp-server community MLKpublic RO
snmp-server community MLKprivate RW
spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree vlan 1-4094 priority 4096
```

```
end
wr
```

---

## L3-SW-Core-4

```
hostname L3-SW-Core-4
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
ip routing
interface range e1/0 , e1/1
channel-group 1 mode active
no shutdown
exit
interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
Exit
```

```
! Note: In topology, L3-SW-Core-4 connects to L3-SW-Core-3 only
! No direct link to ZBP-1 shown. If needed, use e1/2:
! interface e1/2
! no switchport
! ip address 10.20.20.38 255.255.255.252
! no shutdown
! Exit
```

```
interface range e0/0 , e0/1 , e0/2 , e0/3
description *** Trunk to L2-Switches ***
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface vlan 10
ip address 192.168.10.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 20
ip address 10.10.0.3 255.255.0.0
ip helper-address 10.20.20.10
no shutdown
```

```
exit
interface vlan 30
ip address 192.168.30.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 40
ip address 192.168.40.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 50
ip address 10.10.50.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 60
ip address 192.168.60.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 70
ip address 192.168.70.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 80
ip address 192.168.80.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 20
standby 1 ip 10.10.0.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 30
standby 1 ip 192.168.30.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 40
standby 1 ip 192.168.40.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 50
standby 1 ip 10.10.50.1
standby 1 priority 90
standby 1 preempt
```

```
exit
interface vlan 60
standby 1 ip 192.168.60.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 70
standby 1 ip 192.168.70.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 80
standby 1 ip 192.168.80.1
standby 1 priority 90
standby 1 preempt
exit
router ospf 1
router-id 1.1.1.2
network 10.20.20.36 0.0.0.3 area 0
network 10.10.0.0 0.0.255.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.10.50.0 0.0.0.255 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 2
login authentication default
transport input ssh
privilege level 15
exit
line vty 3 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit
ip access-list standard MGMT-ACL
permit 192.168.10.0 0.0.0.255
deny any log
exit
line vty 0 4
```

```
access-class MGMT-ACL in
exit
spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree vlan 1-4094 priority 8192

end
Wr
```

---

## **L3-SW-Core-2**

```
hostname L3-SW-Core-2
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
ip routing
interface range e1/1 , e1/0
channel-group 1 mode active
no shutdown
exit
interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface e1/2
description *** Link to ZBP-2 fa2/0 ***
no switchport
ip address 10.20.20.42 255.255.255.252
no shutdown
exit
interface range e0/0 , e0/1 , e0/2 , e0/3
description *** Trunk to L2-Switches ***
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface vlan 10
ip address 192.168.10.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 20
ip address 10.11.0.2 255.255.0.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 30
ip address 192.168.30.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
```



```
exit
interface vlan 40
ip address 192.168.40.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 50
ip address 10.11.50.2 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 60
ip address 192.168.60.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 70
ip address 192.168.70.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 80
ip address 192.168.80.4 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 20
standby 1 ip 10.11.0.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 30
standby 1 ip 192.168.30.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 40
standby 1 ip 192.168.40.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 50
standby 1 ip 10.11.50.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 60
standby 1 ip 192.168.60.1
standby 1 priority 110
standby 1 preempt
```

```
exit
interface vlan 70
standby 1 ip 192.168.70.1
standby 1 priority 110
standby 1 preempt
exit
interface vlan 80
standby 1 ip 192.168.80.1
standby 1 priority 110
standby 1 preempt
exit
router ospf 1
router-id 2.2.2.1
network 10.20.20.40 0.0.0.3 area 0
network 10.11.0.0 0.0.255.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.11.50.0 0.0.0.255 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 2
login authentication default
transport input ssh
privilege level 15
exit
line vty 3 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit
ip access-list standard MGMT-ACL
permit 192.168.10.0 0.0.0.255
deny any log
exit
line vty 0 4
access-class MGMT-ACL in
exit
spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
```

spanning-tree vlan 1-4094 priority 4096

end  
wr

---

## **L3-SW-Core-1**

```
hostname L3-SW-Core-1
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
ip routing
interface range e1/0 , e1/1
channel-group 1 mode active
no shutdown
exit
interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
Exit
```

! Note: In topology, L3-SW-Core-1 connects to L3-SW-Core-2 only

! No direct link to ZBP-2 shown. If needed, use e1/2:

! interface e1/2

! no switchport

! ip address 10.20.20.46 255.255.255.252

! no shutdown

! exit

```
interface range e0/0 , e0/1 , e0/2 , e0/3
description *** Trunk to L2-Switches ***
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
interface vlan 10
ip address 192.168.10.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 20
ip address 10.11.0.3 255.255.0.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 30
ip address 192.168.30.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
```

```
interface vlan 40
ip address 192.168.40.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 50
ip address 10.11.50.3 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 60
ip address 192.168.60.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 70
ip address 192.168.70.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 80
ip address 192.168.80.5 255.255.255.0
ip helper-address 10.20.20.10
no shutdown
exit
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 20
standby 1 ip 10.11.0.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 30
standby 1 ip 192.168.30.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 40
standby 1 ip 192.168.40.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 50
standby 1 ip 10.11.50.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 60
standby 1 ip 192.168.60.1
standby 1 priority 90
standby 1 preempt
exit
```

```
interface vlan 70
standby 1 ip 192.168.70.1
standby 1 priority 90
standby 1 preempt
exit
interface vlan 80
standby 1 ip 192.168.80.1
standby 1 priority 90
standby 1 preempt
exit
router ospf 1
router-id 2.2.2.2
network 10.20.20.44 0.0.0.3 area 0
network 10.11.0.0 0.0.255.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.11.50.0 0.0.0.255 area 0
exit
aaa new-model
aaa authentication login default local
aaa authorization exec default local
username admin privilege 15 secret 123
enable secret 123
ip domain-name mlkuniversity.com
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 2
login authentication default
transport input ssh
privilege level 15
exit
line vty 3 4
login authentication default
transport input ssh
privilege level 15
exit
line con 0
login authentication default
password 123
exit
ip access-list standard MGMT-ACL
permit 192.168.10.0 0.0.0.255
deny any log
exit
line vty 0 4
access-class MGMT-ACL in
exit
spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree vlan 1-4094 priority 8192
```

```
end  
wr
```

---

## **L2-SW-5 (VLAN 30 )**

```
hostname L2-SW-5  
configure terminal  
vtp domain MLK-University  
vtp mode client  
vtp version 2  
vtp password cisco123  
interface range e0/2 , e0/3  
switchport mode dynamic d  
switchport trunk allowed vlan 10,20,30,40,50,60,70,80  
exit  
interface range e0/0 , e0/1  
switchport mode access  
switchport access vlan 30  
spanning-tree portfast  
spanning-tree bpduguard enable  
exit  
  
end  
wr
```

---

## **L2-SW-6 (VLAN 40 )**

```
hostname L2-SW-6  
configure terminal  
vtp domain MLK-University  
vtp mode client  
vtp version 2  
vtp password cisco123  
interface range e0/2 , e0/3  
switchport mode dynamic d  
switchport trunk allowed vlan 10,20,30,40,50,60,70,80  
exit  
interface range e0/0 , e0/1  
switchport mode access  
switchport access vlan 40  
spanning-tree portfast  
spanning-tree bpduguard enable  
exit  
  
end  
wr
```

---

## **L2-SW-7 (VLAN 60 )**

```
hostname L2-SW-7  
configure terminal  
vtp domain MLK-University  
vtp mode client  
vtp version 2  
vtp password cisco123
```

```
interface range e0/2 , e0/3
switchport mode dynamic d
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
exit
interface range e0/0 , e0/1
switchport mode access
switchport access vlan 60
spanning-tree portfast
spanning-tree bpduguard enable
exit

end
wr
```

---

## **L2-SW-8 (VLAN 80 )**

```
hostname L2-SW-8
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
interface range e0/2 , e0/3
switchport mode dynamic d
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
exit
interface range e0/0 , e0/1
switchport mode access
switchport access vlan 80
spanning-tree portfast
spanning-tree bpduguard enable
exit

end
wr
```

---

## **L2-SW-1 (VLAN 10 )**

```
hostname L2-SW-1
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
interface range e0/2 , e0/3
switchport mode dynamic d
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
exit
interface range e0/0 , e0/1
switchport mode access
switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
Exit
```

end  
wr

---

## **L2-SW-2 (VLAN 20 )**

```
hostname L2-SW-2
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
interface range e0/2 , e0/3
switchport mode dynamic d
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
exit
interface range e0/0 , e0/1
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

end  
wr

---

## **L2-SW-3 (VLAN 50 )**

```
hostname L2-SW-3
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
interface range e0/2 , e0/3
switchport mode dynamic d
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
exit
interface range e0/0 , e0/1
switchport mode access
switchport access vlan 50
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

end  
wr

---

## **L2-SW-4 (VLAN 70 )**

```
hostname L2-SW-4
configure terminal
vtp domain MLK-University
vtp mode client
vtp version 2
vtp password cisco123
interface range e0/2 , e0/3
```



end  
Wr